Sec: SR.IIT_*CO-SC(MODEL-A& APEX) Date:27-12-24
Time: 3HRS GTM-4 Max. Marks: 300

## KEYSHEET

### PHYSICS

| 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| A | B | C | B | D | A | D | D | A | D |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| B | D | C | D | C | A | D | B | D | B |
| 21 | 22 | 23 | 24 | 25 | | | | | |
| 495 | 8 | 4 | 2 | 9 | | | | | |

### CHEMISTRY

| 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 |
|---|---|---|---|---|---|---|---|---|---|
| B | D | D | C | B | B | C | A | B | A |
| 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 |
| D | D | B | B | D | D | B | D | D | C |
| 46 | 47 | 48 | 49 | 50 | | | | | |
| 4 | 4 | 5 | 6 | 32 | | | | | |

### MATHEMATICS

| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
|---|---|---|---|---|---|---|---|---|---|
| B | A | B | A | B | B | C | D | C | C |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| A | A | A | B | D | A | C | B | D | C |
| 71 | 72 | 73 | 74 | 75 | | | | | |
| 40 | 3 | 0 | 30 | –2 | | | | | |



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1. density of cylinder $\rho = \frac{m}{\pi r^2 l}$

$$\frac{\Delta\rho}{\rho}\times 100 = \frac{\Delta m}{m}\times 100 + \frac{2\Delta r}{r}\times 100 + \frac{\Delta l}{r}\times 100$$

$$= 1\% + 2\times 1.5\% + 2\%$$

$$= 6\%$$

2. Power factor $\cos\phi = \frac{R}{Z} = \frac{6}{\sqrt{6^2+8^2}} = \frac{6}{10} = \frac{3}{5}$

3. Total energy, $E_0 = -GMm/2r$

Potential Energy, $U = -GMm/r = 2E_0$

4. Use Bernoulli's theorem & equation of continuity

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho\left(4v_1^2\right) \qquad \left(\therefore v_2 = 2v_1\right)$$

$$P_1 - P_2 = \frac{3}{2}\rho v_1^2$$

$$v_1 = \sqrt{\frac{2\left(P_1 - P_2\right)}{3\rho}} = \sqrt{\frac{2\times 4500}{3\times 750}} = 2m/s$$

5. Net force is by nuclear force & electrostatic force

6. $K_{\max} = \frac{1}{4\pi\in_0}.\frac{2e.Ze}{r_0} = \frac{9\times 10^9\times 2\times 80\times 1.6\times 10^{-19}}{9\times 10^{-14}} ev$

$$K_{\max} = 256\times 10^4 ev$$

7. $\theta = \frac{P_\theta - P_0}{P_{100} - P_0}\times 100$

$$= \frac{1.6-1}{2-1}\times 100 = 60^0 C$$

8. $\left|\overrightarrow{v_c}\right| = \frac{\sqrt{p_1^2 + p_2^2}}{m_1 + m_2} = \frac{\sqrt{8^2+6^2}}{2+3} = \frac{10}{5} = 2m/s$

9. 10 VSD = 9MSD = $9\times 1mm$



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L.C = 1 MSD – 1VSD = 1mm-0.9mm = 0.1 mm=0.01 cm

+ve zero error = 0.04 cm

Measured reading $= 4.1cm + 6 \times 0.01cm = 4.16cm$

$= 4.16 - 0.04 = 4.12cm$

10. $i_0 = \frac{3}{1+2} = 1A$

$i = i_0 \times \frac{R_2}{R_1 + R_2} = 1\left(\frac{6}{3+6}\right) = \frac{2}{3}A$

11. $F = \frac{2IA}{c} \Rightarrow A = \frac{FC}{2I} = \frac{2 \times 10^{-4} \times 3 \times 10^8}{2 \times 300} \times 10^{-4} = \frac{1}{100} m^2$

12. $u = u_1 + u_2$

$\frac{n_1 + n_2}{\gamma_{mix} - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1}$

$\frac{6}{\gamma - 1} = \frac{2}{2/3} + \frac{4}{2/5}$

$\gamma = \frac{19}{13}$

13. $P = \sqrt{2mk}$

$P^1 = \sqrt{2m(4k)} = 2P$

$\frac{P^1 - P}{P} \times 100 = \frac{2P - P}{P} \times 100 = 100\%$

14. Momentum conservation

$O = \frac{m}{4}v + \frac{3m}{4}v' \Rightarrow v' = -\frac{v}{3}$

$k = \frac{1}{2}\frac{m}{4}v^2 + \frac{1}{2}\frac{3m}{4}\frac{v^2}{9} = \frac{mv^2}{6}$

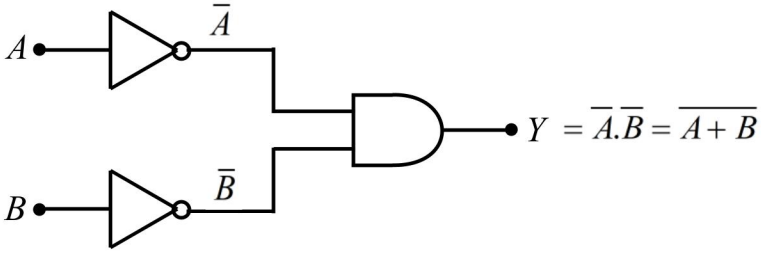
15. $v = r\omega, \quad \omega =$ constant

$\frac{v_1}{v_2} = \frac{r_1}{r_2} \Rightarrow \frac{6}{v_2} = \frac{2}{3} \Rightarrow v_2 = 9m/s$



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16.




17. $\Delta p = \int F dt = \int_{t=0}^{t=3} (4t+1)dt = \left[2t^2 + t\right]_{t=0}^{t=3} = 21 kg\, ms^{-1}$

18. Potentials of A and B is same in contact

$$\frac{1}{4\pi \in_0}\frac{q_1}{r_1} = \frac{1}{4\pi \in_0}\frac{q_2}{r_2}$$

$$\frac{q_1}{q_2} = \frac{r_1}{r_2} = \frac{1}{2} \quad \rightarrow (I)$$

$$q_1 + q_2 = 6\mu c \quad \rightarrow (II)$$

I & II gives $q_1 = 2\mu c$

19. $\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1} = \frac{v_1}{v_2}$

$$\frac{v_1}{v_2} = \frac{\sin 45^0}{\sin 30^0} = \sqrt{2}$$

20. $\lambda = \frac{h}{\sqrt{2mqv}}$

For electron $\lambda = \sqrt{\frac{150}{v}}\,\text{Å} = \sqrt{\frac{150}{300}} = \frac{1}{\sqrt{2}}\,\text{Å}$

21. $\frac{3v}{4l_c} - \frac{2v}{2l_0} = 4$

$$\frac{990}{4l_c} - \frac{330\times 100}{30} = 4$$

$$\frac{990}{4l_c} = 4 + 1100$$



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$$l_c = \frac{990}{1104 \times 4} = \frac{495}{2208}$$

22. $r = \frac{mv}{Bq} \Rightarrow B = \frac{mv}{qr} = 0.2T$

$$v = \frac{E}{B} \Rightarrow E = vB$$

$$E = 4 \times 10^5 \times 0.2 = 8 \times 10^4 v/m$$

23. $W = S\Delta A = S.4\pi R^2 \left( n^{\frac{1}{3}} - 1 \right)$

$$W = 192\pi K \times 10^{-5} J$$

24. $t = \frac{2l}{v}$

Induced emf $e = Bvl$

Heat produced $H = \frac{e^2 t}{R} = \frac{(Bvl)^2}{R} \frac{2l}{v} = \frac{2B^2 vl^3}{R}$

25. $v_{(1,2)} - v_{(0,0)} = -\int \vec{E}.\vec{dr} = \left[ \int_0^1 2xdx + \int_0^2 3y^2 dy \right]$

$$\left[ x^2 \right]_0^1 + \left[ y^3 \right]_0^2$$

$$v_{(1,2)} = 1 - 0 + 8 - 0$$

$$= 9V$$

## CHEMISTRY

26. $K_{sp} = S^2 = 10^{-4} \Rightarrow s = 10^{-2} M$

$\therefore$ Molarity of the saturated solution is $10^{-2}$ M

$H^+$ removes $CO_3^{2-}$ from the solution by forming undissociated $HCO_3^-$ which drives the forward reaction.

$K_{sp}$ is not affected by common ion

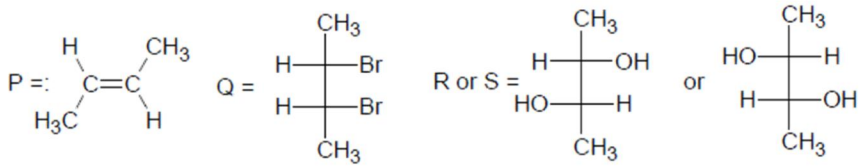
Ionic product ($10^{-6}$) is less than $K_{sp}$ ($10^{-4}$). Hence, no ppt. is formed

27. Nitrate gives $NO_2$ gas with concentrated $H_2SO_4$

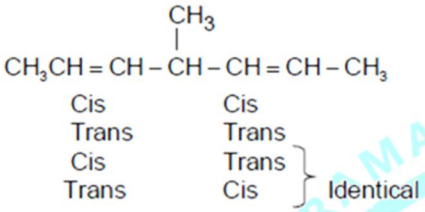
28.



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29. Due to double bond, the geometrical isomers are




When the double bonds are cis and trans, the molecule becomes optically active.

$\therefore$ For trans cis (two optical isomers are possible)

$\therefore$ Total no. of stereoisomers are :

Geometrical (2) + Optical (2) = 4

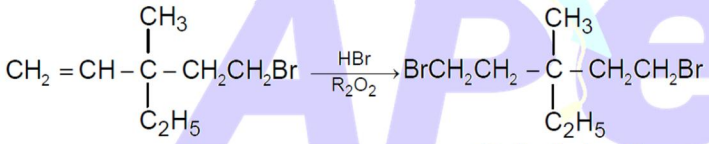
30. $\dfrac{p^o - p_S}{p^o} = \dfrac{2}{8+2} = \dfrac{2}{10}$

31. $NH_4Cl \longrightarrow NH_4^+ + Cl^-$

$NH_4^+ + H_2O \longrightarrow NH_4OH + H^+$

$H^+$ shift the ionization reaction toward backward direction due to common ion effect.

32.




Optically inactive

33. No. of radial nodes = (n – $l$ – 1)

For 4s orbital, it is 3

For 4p orbital, it is 2

For 4d orbital, it is 1

For 4f orbital, it is 0

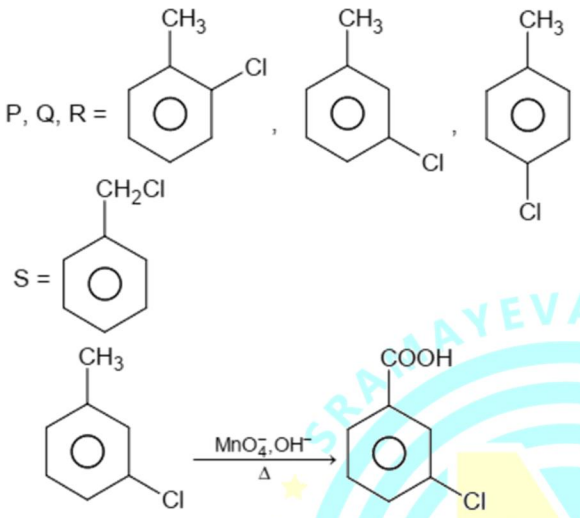
34. Divide the mass% of corresponding elements with their atomic masses and take the simple ratio of the values obtained. Empirical formula is $C_6H_5NO$.

35. Less electronegative is better nucleophile



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36.




37. $E_{Cell} = E^{o}_{Cell} - \frac{0.0591}{n}\log\left[H^{+}\right] = 0 - \frac{0.0591}{1}\log 10^{-4} = -0.0591 \times -4 = 0.2364\ \text{V}$

38. Pyrogallol is 1, 2, 3 trihydroxy benzene

39. In irreversible isothermal compression, heat is transferred from system to surrounding.

$\therefore$ The entropy of surrounding increases.

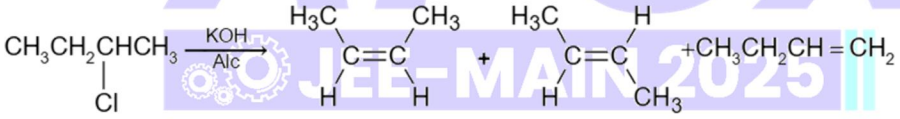
40. $2SO_2 + O_2 \rightarrow 2SO_3$

$\Delta H = \Delta_f H^0(product) - \Delta_f H^0(reactant)$

$= 2\times(-y) - [2(-x) + 0] = -2y + 2x = (2x - 2y)\,Kcal$

41. Conceptual

42.




43. For adiabatic process

$$\%N = \frac{1.4\times 300\times 0.5\times 2}{0.7} = 60\%$$

44. A $\rightarrow$ Aromatic, B $\rightarrow$ Non-aromatic, C $\rightarrow$ Aromatic, D $\rightarrow$ Antiaromatic

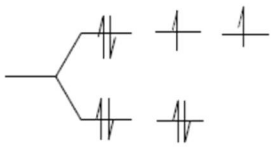
45. Correct order for bond dissociation energy is

| | | | | | |
|---|---|---|---|---|---|
| | $Cl_2$ | > $Br_2$ | > | $F_2$ | > $I_2$ |
| kJ / mol | (242) | (193) | | (159) | (151) |



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46.

$[NiCl_4]^{2-}$ $Ni^{2+}$ (Td)  $n_1 = 2$

$[Ni(CN)_4]^{2-}$ $Ni^{2+}$ (Sq. planar)  $n_2 = 0$

$[Ni(H_2O)_6]^{2+}$ $Ni^{2+}$ (Oh)  $n_3 = 2$

Total number of unpaired electrons = $n_1 + n_2 + n_3 = 4$

47. Conceptual

48. The paramagnetic species are $O_2, O_2^+, B_2, N_2^+, O_2^-$.

49. $31X(g) \longrightarrow Y(g)$

At 30 min, [X] = [Y]

$\therefore 31X(g) \longrightarrow Y(g)$

A – 31x x

$\therefore$ a – 31x = x at t = 30 min

or, $a = 32x \Rightarrow x = \dfrac{a}{32}$

$\therefore 5t_{1/2} = 30 \text{ min} \quad t_{1/2} = 6 \text{min}$

50. $X(x) \rightleftarrows Y(g) + 2Z(g)$

Total equilibrium pressure = 6 atm

$P_Y + P_z = 6$

or $x + 2x = 6 \Rightarrow x = 2$

$p_y = 0.6,\ p_z = 1.2$

$\therefore K_P = (p_y)(p_z)^2 = (2)(4)^2 = 32$



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51. $\int dy = \int 2(x+1)\,dx$

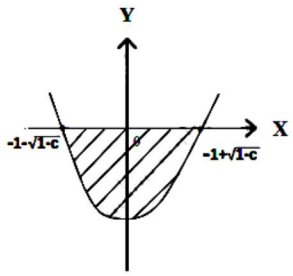
$y = x^2 + 2x + c = (x+1)^2 - (1-c)$

Put $y = 0$

$(x+1)^2 = 1-c$

$x = -1 \pm \sqrt{1-c}$

Required area $= -\int_{-1-\sqrt{1-c}}^{-1+\sqrt{1-c}} \left[(x+1)^2 - (1-c)\right] dx = \frac{8\sqrt{2}}{3}$




Solving $c = -1$

Put $x = 2$ in $y = x^2 + 2x - 1$

$y = 7$

52. $C_1 = (-3, -4),\ r_1 = \sqrt{9+16-16} = 3$

$C_2 = \left(\sqrt{3}-3, \sqrt{6}-4\right),\ r_2 = \sqrt{\left(\sqrt{3}-3\right)^2 + \left(\sqrt{6}-4\right)^2 + k + 6\sqrt{3} + 8\sqrt{6}}$

$= \sqrt{34+k}$

Use $C_1C_2 = |r_1 - r_2|$

$3 = \sqrt{34+k} - 3$

$k = 2$ and $r_2 = 6$

Point of contact P divides $\overline{C_1C_2}$ in the ratio $r_1 : r_2 = 3:6$ externally

53. put $e^x = t > 0$

$(t^2 - 4)(2t-1)(3t-1) = 0$

$t = 2, \frac{1}{2}, \frac{1}{3}$



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$e^{x_1}.e^{x_2}.e^{x_3} = (2)\left(\frac{1}{2}\right)\left(\frac{1}{3}\right) = \frac{1}{3}$

$e^{x_1+x_2+x_3} = \frac{1}{3}$

$\therefore x_1 + x_2 + x_3 = \log_e\left(\frac{1}{3}\right)$

54. $4x^2 + 11x + 6 > 0, -1 \le 4x + 3 \le 1$ and $-1 \le \frac{10x+6}{3} \le 1$

$(4x+3)(x+2) > 0$, $-1 \le x \le -\frac{1}{2}$ and $-\frac{9}{10} \le x \le -\frac{3}{10}$

$x < -2$ or $x > -\frac{3}{4}$

Common values are given by $\left(-\frac{3}{4}, -\frac{1}{2}\right]$

55. Let $Q = \left(\lambda, \frac{3\lambda - 5}{2}\right)$ and $R = (\mu, 14 - 4\mu)$

$$\left(2, -\frac{4}{3}\right) = \left[\frac{1(\mu) + 2(\lambda)}{1+2}, \frac{1(14-4\mu) + 2\left(\frac{3\lambda-5}{2}\right)}{1+2}\right]$$

$$= \left(\frac{\mu + 2\lambda}{3}, \frac{-4\mu + 3\lambda + 9}{3}\right)$$

$\mu + 2\lambda = 6,\ 4\mu - 3\lambda = 13$

$\lambda = 1, \mu = 4$

Equation of line passing through $Q(1, -1), R(4, -2)$ is

$y + 1 = -\frac{1}{3}(x - 1)$

$x + 3y + 2 = 0$

56. $\left((\overline{p} + \overline{q}) + (-2\overline{p} + 3\overline{q})\right) \times \overline{r} = \overline{0}$

$(4\overline{q} - \overline{p}) \times \overline{r} = \overline{0}$

Let $\overline{r} = \lambda(4\overline{q} - \overline{p})$

$= \lambda\left(40\hat{i} - 3\hat{j} + 3\hat{k}\right)$

$2\overline{p} + 3\overline{q} = 41\hat{i} - 5\hat{j} + 5\hat{k}$



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$\lambda(1640+15+15)=1670$

$\lambda=1$

$|\overline{r}|^2=1600+9+9=1618$

57. $\dfrac{x-1}{2}=\dfrac{y-z}{1}=\dfrac{z-1}{2}=k \Rightarrow \dfrac{x-1}{2}=\dfrac{y-1}{3}=\dfrac{z-1}{2}=k$

$\dfrac{x-5}{2}=\dfrac{y-p}{-1}=\dfrac{z-5}{3}=t$

$2k+1=2t+5$

$2k+1=3t+5$

$3k+1=-t+p$

$t=0,\ k=2$

58. $\left|P^{-1}AP-2P^{-1}IP\right|=\left|P^{-1}(A-2I)P\right|$

$=\left|P^{-1}\right|\left|A-2I\right|\left|P\right|=\dfrac{1}{|P|}|A-2I||P|$

$=|A-2I|=\begin{vmatrix}0 & 1 & 2\\ 6 & 0 & 11\\ 3 & 3 & 0\end{vmatrix}=69=3\times 23$

$2+23=26$

59. $f'(x)=5(x+1)(x+3)(x-2)^2=0$

$x=-3,-1,2$

Observe $f(-4), f(-3), f(-1), f(2), f(4)$

$M=392,\ m=-216$

60. $\displaystyle\sum_{r=0}^{50}\frac{{}^{100-r}C_{25}}{100-r}\Rightarrow\sum_{r=0}^{50}\frac{\frac{100-r}{25}\times{}^{99-r}C_{24}}{100-r}$

$\left[{}^{n}C_r=\dfrac{n}{r}\times{}^{n-1}C_{r-1}\right]$

$=\dfrac{1}{25}\displaystyle\sum_{r=0}^{50}{}^{99-r}C_{24}$



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$$\frac{1}{25}\left[{}^{99}C_{24}+{}^{98}C_{24}.....{}^{49}C_{24}\right]$$

$$\frac{1}{25}\left[{}^{100}C_{25}-{}^{49}C_{25}\right]$$

61.

$$f(x)=|\cos x|+|\sin x|$$

$$[f(x)]=1$$

So, $\int_{-3/2}^{3/2} 1dx=[x]_{-3/2}^{3/2}=3$

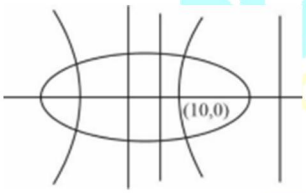
62.

We have to find $\left|(2a_1)^2-(2a_2)^2\right|$

$a_1e_1=a_2e_2=10$

and $e_1=\frac{10}{a_1}\ ;\ e_2=\frac{10}{a_2}$

also $\left|\frac{a_1}{e_1}-\frac{a_2}{e_2}\right|=10\ ;\ x=\frac{a_2}{e_2}\ ;\ x=\frac{a_1}{e_1}$




$$\left|\frac{a_1^2}{10}-\frac{a_2^2}{10}\right|=10\ ,\ \left|a_1^2-a_2^2\right|=100$$

$$\left|4a_1^2-4a_2^2\right|=400$$

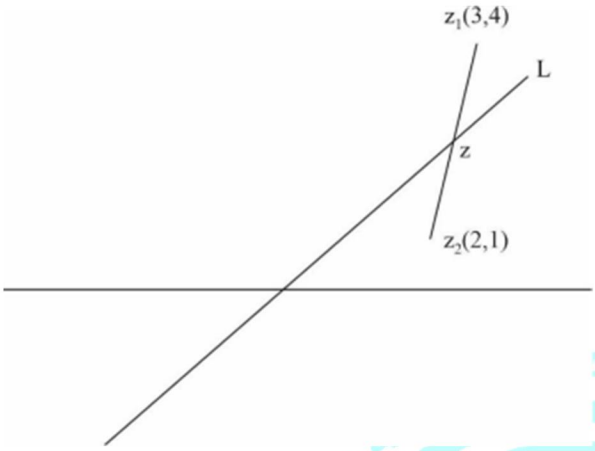
63. Clearly $2\left(z+\bar{z}\right)+3\left(z-\bar{z}\right)i=0$

$2x-3y=0$ (a line)

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z lies on intersection of L and line joining

$z_1$ *and* $z_2$ *i.e.*, $y-1=3(x-2)$

i.e., $3x-y=5$ and $2x-3y=0$

$$7x=15 \quad y=\frac{45}{7}-5=\frac{10}{7}$$

$$x=\frac{15}{7} \;;\; z=x+yi=\frac{15}{7}+\frac{10}{7}i$$

64.

$$x+\sqrt{1+x^2}=\frac{1}{\sqrt{1+y^2}+y}=\sqrt{1+y^2}-y \qquad ...(1)$$

$$y+\sqrt{1+y^2}=\frac{1}{\sqrt{1+x^2}+x}=\sqrt{1+x^2}-x \qquad ...(2)$$

Adding (1) and (2) we get

$x+y=0 \Rightarrow y=-x$

$\frac{dy}{dx}=-1$

65. Let A = families own a Car

B = families own a Scooter

C = families own a Bicycle

We have to find

$$\Rightarrow \quad n(A' \cap B' \cap C') = n(U) - n(A \cup B \cup C)$$

$$= 100 - \left[\sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)\right]$$

$$= 100 - \left[(60+80+40) - 90 + x\right]$$



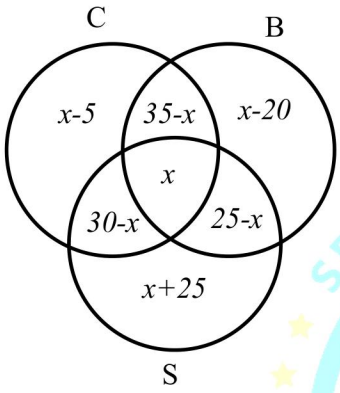
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$=100-[90+x]$

$10-x$

[Here $x=n(A\cap B\cap C)>0$ and $0<x\le 10$]

Also,




$x-20\ge 0$

Hence, no value of $x$.

66. $\left(1^2-2^2\right)+\left(4^2-5^2\right)+\left(7^2-8^2\right)+....\text{up to 30 terms}$

$+\left(3^2+6^2+9^2+....\text{upto 30 terms}\right)$

$-(3+9+15+.....\text{upto 30terms})+3^2\left(1^2+2^2+3^2+....\text{up to 30 terms}\right)$

$-2700+85095$

$=82395.$

67. Multiplying both, get answer $x+y=\pi-2\theta$

68. $(a,0)$ should be on directrix

$-a^2-a=a$

$a=0,-2$

69. Letter $L_4$ is empty

Case 1: if $E_4$ is empty $\Rightarrow L_4$ is not going to $E_4$

$n_1=D(5)=44$

Case 2: if $E_4$ is empty $\Rightarrow L_4$ is going to $E_4$ then

$n_2=D(4)=9$

Total $=44+9=53$



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70. $f(n) = -8 \forall n \in N$ (i.e. independent of n)

71. We have $\frac{dy}{dx} = 20\left(1600 - y^2\right)$

$$\Rightarrow \int \frac{dy}{(40)^2 - y^2} = 20\int dx$$

$$\Rightarrow \frac{1}{80}\ln\frac{40+y}{40-y} = 20x + C'$$

or $\ln\frac{40+y}{40-y} = 1600x + C$

$$\Rightarrow \frac{40+y}{40-y} = \frac{ke1600x}{1}, \text{ where } k = e^C\,(let)$$

$$\Rightarrow \frac{2y}{80} = \frac{ke^{1600x} - 1}{ke^{1600x} + 1} \text{ (using componendo and dividend)}$$

$$\therefore \quad \lim_{x\to\infty} y = 40\lim_{x\to\infty}\left[\frac{k - e^{-1600x}}{k + e^{-1600x}}\right] = 40$$

72.

$$\left(\frac{3}{5} + \frac{2}{5}\right)^7$$

we here to get the greatest term.

Let $T_{r+1}$ is greatest term

$$\left(\frac{3}{5}\right)^7\left[1 + \frac{2}{3}\right]^7 \quad x = \frac{2}{3}$$

$$m = \frac{|x|(n+1)}{|x|+1} = \frac{2/3(7+1)}{\frac{2}{3}+1} = \frac{16}{5}$$

$$r = [m] = \left[\frac{16}{5}\right] = 3$$

73. Domain of

$\sin^{-1}x + \cos^{-1}x + \tan^{-1}x$ is $[-1, 1]$

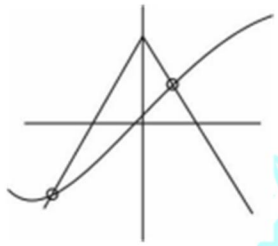


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So range is

$$\left[\frac{\pi}{2}-\frac{\pi}{4},\frac{\pi}{2}+\frac{\pi}{4}\right]=\left[\frac{\pi}{4},\frac{3\pi}{4}\right]$$

$$b=\frac{3\pi}{4} \Rightarrow \quad [b]=2$$



Now no. of solutions of $1-|x|=\tan^{-1}x$

2 solutions.

74. $$\bar{x}=\frac{\sum x_i}{100}=0 \,\&\, \frac{\sum|x_i-\bar{x}|}{100}=5 \Rightarrow \sum_{1\le i<j\le 100}|x_i x_j|=(500)^2$$

$$\Rightarrow \frac{\sum x_i^2}{100}=\frac{(500)^2-2\sum|x_i x_j|}{100}=2500-1600$$

$$\text{S.D}=\sqrt{\frac{\sum(X_i-\bar{X})^2}{100}}=\sqrt{900}=30$$

75. $$\text{RHD}=\lim_{n\to 0^+}\frac{ae^{1/n}+3e^{-1/n}}{(a+2)e^{1/n}-e^{-1/n}}=\frac{a}{a+2}$$

$$\text{LHD}=\lim_{h\to 0^+}\frac{ae^{1/n}+3e^{1/n}}{(a+2)e^{1/n}-e^{1/n}}=\frac{a+3}{a+1}$$

$$a=-3/2$$



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